# Chengxi Yang

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## Education

**University of California, Berkeley** — Berkeley, CA, USA
*Ph.D. in Computer Science* — *Aug. 2026 - Present*
- Advisor: Prof. Alvin Cheung

**Shanghai Jiao Tong University** — Shanghai, China
*Bachelor in Computer Science* — *Sep. 2022 - Jun. 2026*
- A member of ACM Honors Class, an elite CS program in SJTU
- **GPA**: **92.7**/100, Ranking: 1/29, 22 A+ courses

## Research Interest

My research interest mainly lies in **programming languages** and **formal methods**. I'm also interested in the intersection of AI and PL/FM.

## Research Experience

**University of Wisconsin-Madison** — Madison, WI, USA
*Research Intern at madPL, co-advised by Prof. Thomas Reps and Prof. Tej Chajed* — *Jun. 2025 - Dec. 2025*
Developed a formal theory of **convergence detection** for incremental recursive computation, with key results mechanized in **Lean**.

**Shanghai Jiao Tong University** — Shanghai, China
*Research Assistant, advised by Prof. Qinxiang Cao* — *Jun. 2024 - Jun. 2026*
Developed methods and **Rocq** infrastructure for algorithm verification, and contributed to the **QCP verification tool** and its VS Code extension.

## Publications and Preprints

**Fixing the Fixpoint: A Formal Theory of Convergence Detection for Incremental Recursive Computation**
***Chengxi Yang***, *Tej Chajed, Thomas Reps*
Major revision at OOPSLA 2026; revised version submitted to OOPSLA 2027. [arXiv]

**Assertions for Free: Transferring Invariants from Algorithm to Implementation Proofs (Functional Pearl)**
*Shushu Wu,* ***Chengxi Yang****, Xiwei Wu, Qinxiang Cao*
In ICFP 2026. [DOI]

**QCP: A Practical Separation Logic-based C Program Verification Tool**
*Xiwei Wu, Yueyang Feng*, Xiaoyang Lu*, Tianchuan Lin, Kan Liu, Zhiyi Wang, Shushu Wu, Lihan Xie,* ***Chengxi Yang****, Hongyi Zhong, Naijun Zhan, Zhenjiang Hu, Qinxiang Cao*
In TASE 2026. [arXiv] [DOI]

**Intuitive Verification of Sequential Programs Using Hybrid Reasoning**
*Shushu Wu, Xiwei Wu,* ***Chengxi Yang****, Qinxiang Cao*
In TASE 2026. [DOI]

**A Formal Framework for Naturally Specifying and Verifying Sequential Algorithms**
***Chengxi Yang*****, Shushu Wu*, Qinxiang Cao*
In Theoretical Aspects of Software Engineering (TASE) 2025. [arXiv] [DOI]

## Honors & Awards

**Scholarship**
- **Jun Ye and Neil Shen Zhiyuan Outstanding Scholarship**, 2026
- **OPPO "Benfen" Scholarship** (first-year winner, about 30 winners in SJTU), 2025
- **Fan Xuji Scholarship** (about 15 winners each year in SJTU), 2024
- **Zhiyuan Honorary Scholarship** (awarded to top 10% students in SJTU annually), 2022-2025

## Selected Projects

- **Mx* Compiler**: A Compiler from Mx* language (a C & Java like language) to RISC-V RV32I Assembly implemented in Java, featuring linear scan register allocation, SSA transformation and various optimization.
- **LTL Model Checker**: A Linear Temporal Logic (LTL) model checker implemented in Rust.
- **RISC-V CPU**: A Tomasulo-based RISC-V RV32I CPU designed in Verilog RTL, featuring instruction cache and branch predictor. It can be run on FPGA board.
- **Acore**: A toy RISC-V kernel implemented in Rust that can be run on QEMU, featuring remote-procedure-call mechanism, virtual memory and preemptive multi-process scheduling.

## Teaching

- **TA for Programming(C++)**. I gave a lecture on introduction to computer systems, designed a new course project and a problem in the final exam.
- **TA for Principles and Practice of Computer Algorithms**. I designed a new lab project on formal verification in Coq.
- **TA for Mathematical Logic**. I answered students' questions, graded their homework and wrote sample solutions for them.
- **TA for Program Verification**. I maintained the course git repository and participated in the design of the final project.

## Skills

**Natural Languages**: Chinese (native), English (fluent).
**Programming Languages**: Rocq (Coq), Lean, Rust, C/C++, Java, Python, Verilog, Go.